SENTHIL KUMAR, APPLIED AI ENGINEER

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- 🧢 🐢 The hats I wear as an Applied AI Engineer:
  - A Generative AI Engineer: Build agentic apps and LLM-powered products
  - An ML Engineer: Leverage self-hosted deep learning for inferencing in NLP applications
  - A Cloud Engineer: Architect & deploy backend AI systems in AWS

- ✍️ Passionate learner and technical writer: 40+ posts at senthilkumarm1901.github.io and 10+ posts on Medium
  - Example blog posts: ❖(1)LangGraph, Strands, and Hermes Comparison; ❖(2)Single LLMs to Agentic AI: GenAI Evolution Explained; ❖(3)Git: TimeTravel Masterclass; ❖(4)Leveraging LLMs in the Cloud

WORK EXPERIENCE

Lead ML Engineer, Toyota Connected India **Jul 2022 – Present**

- Build production-grade LLM-workflows, RAGs and Agentic GenAI applications
- Design event-driven serverless systems on AWS (Lambda, API Gateway, DynamoDB, etc.)
- Lead a group of BE & FE engineers in AI & cloud projects
- Published 4 articles in TCIN Medium channel. Delivered a Talk on Serverless Python App in PyCon2023

Lead Data Scientist, Ford, Analytics Team **May 2018 – Jun 2022**

- A hands-on **data science developer** who co-developed NLP pipelines with experts from the US team
- Leveraged open-source models for inferencing in dockerized NLP applications

Assistant Manager, LatentView Analytics (LV) **Apr 2014 – Apr 2018**

- Roles:**NLP Data Scientist** + **Project Delivery Manager**; led a team of 8+ members

Lead Analyst, Beroe Inc **Jul 2010 – Dec 2013**

- Produced Market Research reports on how to procure indirect spend categories

PERFORMANCE AWARDS

- **Toyota** – Employee of the Month (Sep‘23); 2nd place, AWS India Manufacturing Hackathon (‘24); Team Winner, DriveDesign (‘25 & ‘26), AWS Game Day Security Event (‘26), Hackcelerate (‘23 & ‘26)
- **Ford** – Asia-Pacific Recognition Award (May 2019)
- **LatentView** – Encore Award (Jul–Sep 2016 Quarter)
- **Beroe** – Knowledge Contributor Award (twice, 2013)

EDUCATION

- Bachelor’s in Engg - Electronics - 8.6 CGPA | Madras Institute of Technology, 2006 - 2010
- State Topper in 2006 TN Engineering Entrance Exam (Physical Science-50/50); 12th Grade-95%, 10th Grade-92%

TECHNICAL UPSKILLING

- A 3-Course Specialization on AWS GenAI & Bedrock, Coursera-AWS, Jan‘26
- Modern AI Agents, O’Reilly, Aug‘25 | Building Apps with Anthropic’s MCP, DeepLearning.AI, Jun‘25
- A Proctored Exam on Git, Linux Foundation, Jan‘25
- DevOps on AWS Specialization(4 courses), Coursera-AWS, Oct‘24
- Generative AI with LLMs, Coursera-Deeplearning.ai, Nov‘23
- DeepLearning Specialization(5 courses), Coursera-Deeplearning.ai, Dec‘18 - May‘19

Skills

- **Languages**: Python, SQL(basics), Markdown, Bash/Shell Scripting
- **Python Libraries**: Boto3, LangGraph, Strands, FastAPI, PyTorch, Transformers, and SpaCy
- **Tools**: Git, AWS CLI/Serverless/Bedrock, Docker, Python Env Management, Agentic Engineering

- 👀 **Curious what I have been cooking lately?** 🔗 LinkedIn Featured Section

KEY PROJECTS

Agentic AI Platform | Enterprise Productivity | 2026 | Ongoing

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User ↔ React–Tauri–Rust App: [Agent Loop ↔ LLM API | Tools | DB]
```

- **Impact**: Building a desktop-first Agentic AI Platform letting non-developers create, deploy, and monitor AI agents safely
- **Architecture**: React UI → Tauri IPC → Rust BE
  - Agent Loop inspired by hermes agent loop; architecture similar to Andrew Ng's OpenWorker
- **Tech Stack**: Rust | Tauri 2 | React + TypeScript + Vite | SQLite | Bedrock · Azure OpenAI · Ollama

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GenAI Workflow Platform | Enterprise Productivity | 2024

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Foundation to GenAI apps: YAML Spec → WorkFlow Engine → LLM
Chaining/WorkFlow → Result
```

- **Impact**: Led development of a Serverless GenAI platform on which GenAI apps were built
- **Architecture**: Designed an LLM-chaining workflow engine driven by clear YAML files
- **Tech Stack**: Python | YAML | AWS (Lambda · S3 · Bedrock) | LLM Gateway (Bedrock + Azure OpenAI)

Restricted-Output Conversational AI Agent | Customer Service Automation | 2025–26

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User query → (FSM L1) "Decides WHAT" → (L2) "Does HOW" → LLM & Tools →
Templated Responses
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- **Impact**: Built an explainable, template-controlled customer-service chatbot validated across three concurrent use cases (service booking, vehicle recall check, new car sales) on one orchestration backbone.
- **Architecture**: Designed a two-layer Finite State Machine system (FSM)
- **Tech Stack**: Client: React | API: Node.js | DB: MongoDB | BE: Python & YAML-declared FSM | LLM: OpenAI

PolicyBot: Agentic RAG Chatbot for Enterprise | HR · IT · InfoSec | 2025

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Employee Query → RAG Agent → Tool Call (Vector DB OR MCP Tool) → RAG
Agent Thinks:Confident? –Yes→ Compose Source–cited Answer | –No→
Rephrased Query to RAG Agent
```

- **Impact**: Delivered a role-aware, multilingual policy assistant with *every chunk source-cited* for traceability
- **Architecture**: Architected an agentic RAG system in AWS
- **Highlight**: Moved from Bedrock managed agent orchestration (`agent–runtime`) to self-orchestrated (`bedrock agentcore`) RAG for finer control over planning, prompts, and cost.
- **Tech Stack**: AWS Bedrock | AWS OpenSearch | MCP | OpenAPI Schema | Amazon RDS

Call Center Analytics Dashboard | Contact Center Intelligence | 2024

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audio → (1) diarized turn–level transcript → (2) Prompt–configurable
metric extraction → persisted insights → dashboard
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- **Impact**: Architected a speech-to-insight pipeline turning raw contact-center audio into agent-level performance metrics on a dashboard
- **Highlight**: Migrated from open-source PoC for diarization + STT and metric extraction to serverless managed APIs after cost-usage analysis
- **Tech Stack**: AWS Transcribe (diarization + STT) | LLM API (metric extraction) | AWS (Lambda · S3) | React UI
